The vertex / MPM interface: particle-to-surface contact

`prototype/ecm/test_03_mesh_contact.py` · `log/okuda_ECM/03_mesh_contact` · 10 August 2026

1. The problem, and why the obvious scheme does not fit

The epithelium is a triangulated surface with no mass; the matrix and the basement membrane are MPM continua on a background grid. Every mechanism between them — adhesion, contact, load transmission — has to cross that boundary, and in this prototype it never has: the epithelium is a recorded surface that pushes and is never pushed back.

Outside biology the problem is standard, and it is called FEM − MPM coupling. The natural scheme, *grid-node* coupling (CFEMP, Lian et al. 2011), resolves contact by comparing the two bodies' velocities at grid nodes they share, and it requires the mesh and the grid to be *comparable in size*. Ours are not. A cell is $0.73\,\Delta x$ and the basement membrane $0.1\,\Delta x$, so both bodies fall inside one cell, the grid hands them a single mass-weighted velocity, and they are welded: this is exactly what runs 142–151 measured ten times over. *Particle-to-surface* coupling (ICFEMP, Chen et al. 2015) removes the restriction by testing a particle against a mesh *face* rather than against a node, and that is what is implemented here.

2. The interface, in equations

For particle p under face f with outward normal $\hat{\mathbf{n}}$, let $\mathbf{w}$ be the barycentric weights of p's projection on f and $z_f = \sum_i w_i z_i$ the interpolated surface. The penetration, the normal penalty, and the regularised Coulomb friction are

$$d_p = \big[(\mathbf{x}_p - \mathbf{x}_f)\cdot\hat{\mathbf{n}}\big]_+, \qquad \mathbf{f}_p^n = -\,k\,d_p\,\hat{\mathbf{n}}, \tag{1}$$

$$\Delta\mathbf{v}_t = \big(\mathbf{v}_p - \textstyle\sum_i w_i\mathbf{v}_i\big)_{\perp\hat{\mathbf{n}}}, \qquad \mathbf{f}_p^t = -\min\!\Big(\mu\,k d_p,\ \mu\,k d_p\tfrac{|\Delta\mathbf{v}_t|}{\varepsilon}\Big)\frac{\Delta\mathbf{v}_t}{|\Delta\mathbf{v}_t|}, \tag{2}$$

and the reaction is distributed to the face's vertices by the same weights, which is what makes the exchange conservative by construction:

$$\mathbf{f}_i \;=\; -\,w_i\,\big(\mathbf{f}_p^n + \mathbf{f}_p^t\big), \qquad \text{so} \qquad \sum_p \mathbf{f}_p \;+\; \sum_i \mathbf{f}_i \;=\; \mathbf{0}. \tag{3}$$

ε is a slip velocity, not a fudge: it is the speed below which a contact is treated as stuck, so a resting contact does not chatter.

The one constraint that is not optional. The contact force reaches the particle as a body force, integrated at Δt, so it is stable only while $\Delta t\sqrt{k/m} < 1$, i.e.

$$k \;<\; k_{\max} = \frac{m}{\Delta t^2}, \qquad m = \rho\,\Delta x^3/\mathrm{ppc} = 9.5\times10^{-7}, \quad k_{\max} = 95. \tag{4}$$

The first version of this rig used $k = 3\times10^4$, three hundred times past it. The stiffness is therefore written as a *fraction* of $k_{\max}$ — here 5%, so $k = 0.238$ — which is the same discipline the flat sheet needed after it NaN'd at $k = 6\times10^3$.

3. As a Plexus2 container

The contact is a *relation*, so it is an edge set and not a force field on a body:

object	kind	carries
`contact`	edge set, `pre: mpm_particle`, `post: vertex`	d_p, $\mathbf{w}$, bound flag
`mesh_contact`	**Lateral** on `contact`	returns a delta to *both* endpoints
`contact_rebuild`	**Rewire** on `contact`	which particle is under which face

Two placements matter. `mesh_contact` returns `{mpm_particle: f_p, vertex: f_i}` from one call, so Eq. (3) cannot be half-implemented — which is precisely how the spheroid's adhesion lost its reaction. And it belongs *inside* the `substep_dt` block, recomputed at the substep positions: at frame level a stiff contact is integrated at Δt_{frame} and the ceiling in Eq. (4) falls by $(\Delta t_{\text{frame}}/\Delta t_{\text{sub}})^2 = 400$.

4. The rig, and what it measures

A flat patch of 17×17 vertices with four-neighbour springs, pressed at a prescribed speed into a block of 81,202 MPM particles; mesh cell $1.68\,\Delta x$, no gravity. Three variants, because one of them alone cannot separate the interface from its boundary conditions: the block free, the block trapped against a rigid floor, and the same with a hole cut in the surface. The floor is a *grid* boundary condition — the downward component of the grid velocity is zeroed below the plane, which is how MPM states an obstacle — and the hole is a real hole: 21 vertices inside $r = 0.06$ are removed, the 52 springs touching them go with them, and every quad with a missing corner stops being a contact face, so the material under it has nowhere to go but through.

run		momentum residual	penetration	slip, $\mu = 0.4$ / $\mu = 0$	compression
`03`	free block	1.5×10^{-7}	0.67 cells	—	—
`03b`	rigid floor	1.6×10^{-7}	1.47 cells	0.069 / 0.368	15.0%
`03c`	floor + hole	1.4×10^{-7}	1.56 cells	0.088 / 0.344	14.4%

The momentum residual is float32 machine precision in all three, so the reaction of Eq. (3) is really returned — the test a one-way coupling fails by construction, and the one the spheroid's adhesion would fail today. **Friction changes the slip four- to fivefold**, so the contact is not MPM's automatic weld wearing a coefficient; the free-block run cannot say this, because its slip is measured over the frames that have a contact and it lifts off before enough of them accumulate. **Penetration is bounded but is one to one and a half cells**, which is a lot: the fix is a stiffer penalty with a smaller substep, or barrier contact. And the floor is what turns an indentation into a confined compression — with nowhere to go the block loses 15% of its height, and the hole gives that back slightly (14.4%) because material escapes through it.

The particles are coloured by $|J - 1|$, the volumetric strain, on one 99th-percentile scale taken over the whole run. That scale cannot be known before the run, so the renderer steps the physics once, keeps the drawn frames, and sets the scale from them — the alternative, a value guessed in advance, is what made the falling-block movie a solid saturated colour.

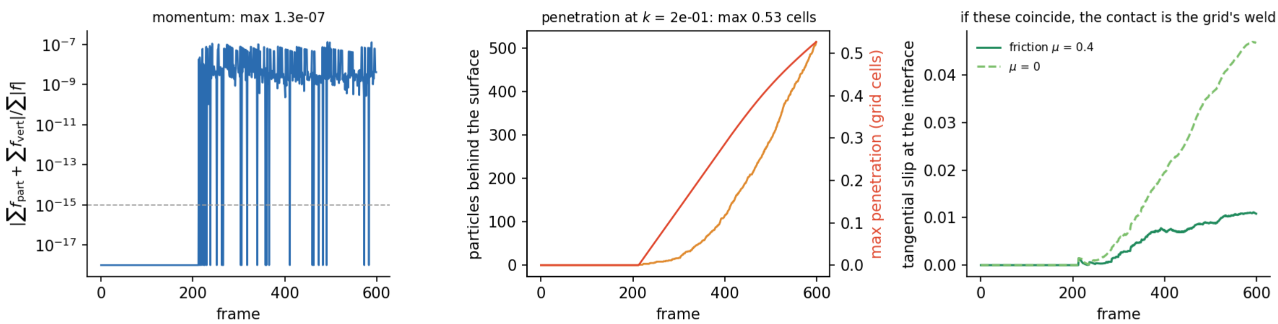

Figure 1: Run `03`. Left: the momentum residual, which is the test that a one-way coupling fails by construction. Middle: how many particles are behind the surface and how deep, in grid cells. Right: tangential slip with friction and without — if these two coincided, the "contact" would be MPM's automatic weld wearing a friction coefficient.

Two failures the rig produced before it produced a result, both worth keeping. Under gravity the block *fell out from under the patch* — its floor is twenty cells below its underside, so it was in free fall and the two bodies never met (block top $0.5800 \to 0.5728$ while the patch went $0.5850 \to 0.5778$). And a patch of independent vertices is not a mesh: pressing its rim left the interior behind, so the lattice needed edge springs before a press could load anything.

5. What this does and does not license

It licenses the interface: a surface and a continuum can exchange force conservatively at a resolution where grid-node coupling cannot, with slip that is a material property rather than a numerical accident, and it survives an obstacle behind the material and a hole in the surface without either being special-cased. It does not license the geometry — the patch is flat and axis-aligned, so face lookup is $O(1)$ and a real mesh needs a BVH — and it does not license the accuracy: half a grid cell of penetration is a lot, and the fix is either a stiffer penalty with a smaller substep or the barrier formulation (BFEMP, Li et al. 2022), which forbids penetration outright at the cost of an implicit solve. The next question is whether the same operator survives a *curved, moving* mesh, which is the spheroid.